

Feathers

Birds are the only animals that have feathers and without them, they would be unable to fly. However, feathers have many other important uses. They help keep birds warm and dry, and they can also play a vital part in courtship displays and in camouflage. Feathers are made of keratin—the same substance that forms hair—and they grow from a bird's skin. Birds can have more than 25,000 feathers, from the soft down feathers covering their bodies, to the long flight feathers in their wings and tails.



*Contour feathers
streamline
bird's body shape*

*Covert feathers smooth
airflow over wing*

*Primary flight
feathers help bird steer*

*Blade, or vane, of flight
feather pushes against air
when wing beats down*

Flight formation

Speeding over the Amazon Rain forest, two red-and-green macaws show off their dazzling plumage. Like all flying birds, they get their smooth body outline from contour feathers, which overlap to create a streamlined shape. The bird's flight feathers are arranged in precise rows on the wings. The longest feathers, known as primaries, lie at the outer edge of each wing. Smaller feathers, called coverts, are arranged over the base of the flight feathers, helping to create lift as the wings move through the air.

Graylag goose



Bath time

Water literally rolls off a duck's back—and off this goose's, too. This happens because contour feathers have microscopic scales that form a water-repellent surface, just like some artificial fabrics. Birds also use their beaks to waterproof their feathers by covering them with oils from their preen gland. Cormorants are the only birds that soak up water when they dive to catch fish. Once back on land, they open up their wings to dry the feathers.